CE323/CE860 Advanced Embedded Systems Design Assignment

The assignment is worth 30% of the total mark for this module. This assignment is to design a Smart Access Control System based on the mbed NXP LPC1768 development board.

The assignment is based on the ARM mbed NXP LPC1768 microcontroller. Some of the details about the mbed will be provided in the **lecture slides**, but full details can be found from the website (Website link). The extension board was developed by the lab technician. The board details can be found from the module website. In the following, a brief description on online compiler, UART link to PC, and the extension board is provided first. Some exercises related to the assignment are then provided to practice. These exercises are helpful for you to better solve the two assignments.

1. The ARM mbed and Online compiler

Each student will be assigned with an mbed and its extension board. The mbed uses an online compiler tool (arm KEIL studio) to edit and compile the source code. You have to create an mbed user account in order to access to the mbed website and resource (use this link to sign up). You must search and select "mbed LPC1768" from the search section.

After log in your account, you can select the compiler menu to go to the workspace. Once going to the workspace, you are able to create new project (from File menu), and edit and compile the source code. The generated bin file can be moved into the mbed folder to execute. The mbed starts to run the latest the bin file once you reset it. The details of using the online compiler can be found here (link).

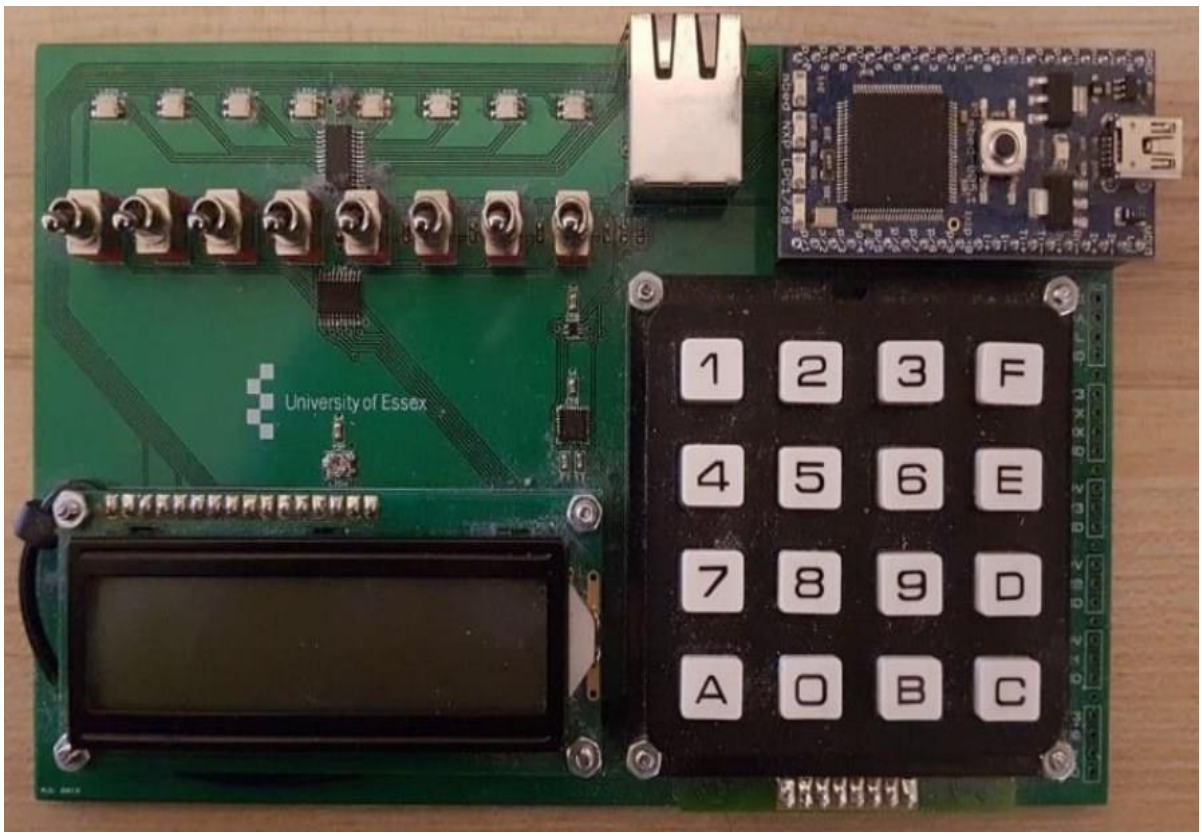

Fig. 1 Physical extension board

2. **Smart Access Control System**

The Smart Access Control System specification is given below. The sensors mentioned below will be replaced with the switches in the virtual extension board and the access indicators are replaced with LEDs in your design. Based on the specification, you need to develop a formal specification using a graphical representation (preferably Unified Modeling Language (UML)). The UML specification should include structure diagrams (class diagrams and/or object diagrams), and behaviour diagrams (state machine diagram, sequence diagram). The formal specification should be written in a report. Then you must write a C/C++ program to implement the design of the home alarm system. You also need to make a demonstration in the lab.

3. The report:

The report should be in PDF and must include a cover page with the module name, student name and ID. The source code should be included in the report. The maximum length of the report is of **5 pages for CE323 Students and 10 pages for CE860 students (without appendix)**.

The report should include:

- **Requirement form**
- **UML graphical representation of the system**
  - class diagrams
  - state machine diagrams
  - sequence diagrams

When necessary, you should provide explanation. **Please note that this is the open end and student-oriented part of the project! the diagrams will not be taught in the class, and you need to self-study through internet to learn them.**

4. Source code

The parts of code you are developing in the labs can be used in your program. You must complete the program and deliver required functionalities.
Full source code should be provided in the report as an appendix.
Full source code should also be provided as a separate document, which can be run in the ARM mbed NXP LPC1768 .

An important note for CE860 Students: the program and functionality are similar. However, higher level of technical explanations is expected from CE860 students. Therefore, the length of the report is considered to be 10 pages.

5. **Submission**:

The report should be submitted through the online coursework submission system (FASER). You should submit a single pdf file called "CE323 Assignment 2 (your name)" for CE323 students and "CE860 Assignment 2 (your name)" for CE860 students. **Full source code should also be uploaded as a separate document, which can be run in the ARM mbed NXP LPC1768** .

University of Essex

Assessment criteria:

- Following the rules mentioned in this document for the submission and preparation of the assignment 10%
- Class diagram 10%
- State machine diagram 10%
- Sequence diagram 10%
- Correctness of program 30%
- Quality and Clarity of program 30%

6. Late submission and plagiarism:

This assignment is to be done individually, i.e. whatever you submit must be your own individual work. Any software or any other materials that you use in this assignment, whether previously published or not, must be referred to and properly acknowledged. Please be aware that the module supervisor may ask students for an interview to explain their submitted work. Please refer to the Undergraduate Students' Handbook for details of the school policy

regarding late submission and University regulations regarding plagiarism:

https://www.essex.ac.uk/student/academic-skills/academic-integrity

https://www.essex.ac.uk/student/exams-and-coursework#academic-offences

Smart Access Control System

1. Introduction to Access Control Systems

A Smart Access Control System manages entry and exit to secure facilities such as office buildings, laboratories, or server rooms (in this link you can find an of an access control system). Your task is to implement a control unit and the functionality behind its user interface. To complete this task you must first understand the specification of the external parts provided and the required behaviour of the overall system.

A conventional access control system normally contains:

- Central control unit
- Keypad entry point / user interface
- Access indicators (LEDs for granted/denied)
- Door sensors such as: magnetic contacts, motion sensors (PIRs), occupancy sensors.

In the case of the sensors, they all contain switches that are "closed" circuit under normal circumstances. This type of switch is known as a Normally Closed (NC) switch, the converse is known as Normally Open (NO). Magnetic contacts (The principle of Magnetic door contact switch refer to Link) are used to detect when doors and windows are opened. When a door is opened a magnet is moved away from the reed switch which causes the switch contacts to open. Likewise, when a movement sensor is triggered or a pressure mat depressed the circuit is broken (the switch is opened), see Figure 1.

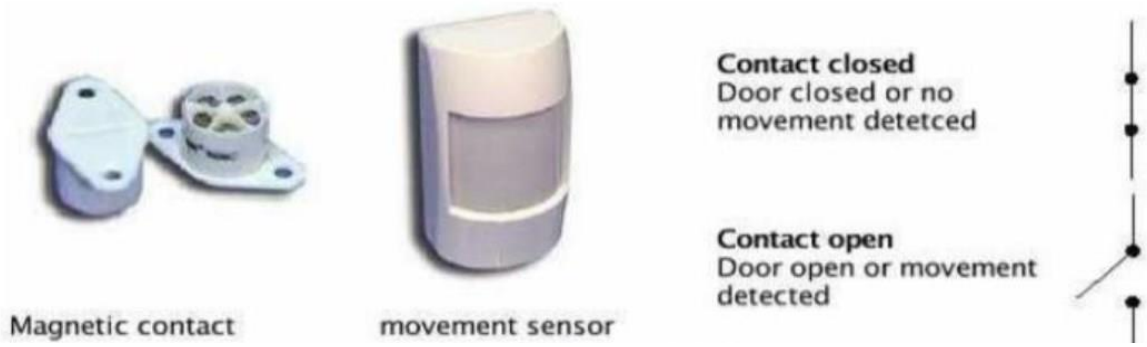

Figure 1: Access Control System Sensors

In this assignment:
• Sensors are emulated using using normally closed push to break switches.
• Access granted/denied indicators are shown using LEDs
• A five-digit PIN code system is used for authentication
• The system tracks access attempts and manages security zones

The switches can emulate sensors in the following way: the opening of a contact can be detected by using one of the microcontroller's parallel port inputs, connected to the top of the switch whose lower end is connected to GND. The top of the switch is also connected through a resistor to 3.3v (V+). When the switch is closed the port, input is connected to 0v (GND), and it will be pulled-up to 3.3v (V+) by the resistor when the sensor's switch opens. Therefore, the microcontroller input is logic low (0v) when a door is closed / there is no movement, and logic high (3.3v) when the sensor is activated.
Normal access control systems include multiple sensors located in various locations, each location can have its own circuit which allows identification of the triggered switch, i.e. it lets the user know

when a sensor is activated. Each sensor circuit is known as a zone, and zones can have different behaviour. The system includes only four sensors.

2. Functional Specification

The Smart Access Control System has SEVEN states of operation: idle, code_entry, access_granted, access_denied, lockdown, monitoring, and admin. Initially it is in the idle state.

- **Idle State:**
  - In the idle state, the system waits for user interaction.
  - The LCD displays "Access Control" on line 1 and "Enter Code" on line 2.
  - All sensor LEDs should be OFF.
  - When the first digit is entered, transition to code_entry state.
  - If any sensor is triggered while in this state, transition to monitoring state.

- **Code Entry State:**
  - User has 30 seconds to enter a five-digit code.
  - LCD line 1 displays "Enter Code"
  - LCD line 2 displays "Code: * _ _ _ _" with "_" replaced as digits are entered.
  - Pressing "C" deletes the last entered digit.
  - After entering 5 digits, LCD line 2 displays "Press B to Submit"
  - Pressing "B" with correct code → access_granted state
  - Pressing "B" with incorrect code → access_denied state
  - If 30 seconds elapse without submission → idle state
  - Three consecutive failed attempts → lockdown state
  - If any sensor is triggered while in this state, transition to monitoring state.

- **Access Granted State:**
  - Green LED (access granted indicator) turns ON for 10 seconds.
  - LCD line 1 displays "Access Granted"
  - LCD line 2 displays "Welcome User"
  - Door unlock signal activated (simulated by a specific LED).
  - After 10 seconds, if door sensor shows door is closed → idle state
  - If door sensor shows door opened and closed within 15 seconds → idle state
  - If door remains open beyond 15 seconds → monitoring state (door held open alarm)
  - If any sensor is triggered while in this state, transition to monitoring state.

- **Access Denied State:**
  - Red LED (access denied indicator) blinks for 5 seconds.
  - LCD line 1 displays "Access Denied"
  - LCD line 2 displays "Attempt X of 3" (X = current failed attempt)
  - After 5 seconds → idle state
  - If this is the 3rd consecutive failed attempt → lockdown state
  - If any sensor is triggered while in this state, transition to monitoring state.

- **Lockdown State:**
  - Triggered after 3 consecutive failed access attempts.
  - Red LED flashes rapidly.
  - LCD line 1 displays "LOCKDOWN"
  - LCD line 2 displays "Wait 2 min"

- System locked for 2 minutes - no code entry accepted.
- After 2 minutes → idle state (failed attempt counter resets)
- Admin code (special 5-digit code, different from user code) can override and go to admin state
- If any sensor is triggered while in this state, transition to monitoring state.

• **Monitoring State:**
- Entered when sensors detect unauthorized activity.
- Amber LED (monitoring indicator) blinks.
- LCD line 1 displays "Monitoring Active"
- LCD line 2 displays sensor status: "Sensor: X" (X = sensor number 1-4)
- Correct user code + B → idle state
- Admin code + B → admin state
- If sensors inactive for 30 seconds → idle state

• **Admin State:**
- Entered with special admin code (different from user access code).
- LCD line 1 displays "Admin Mode"
- LCD line 2 displays "Press A: Reset"
- Pressing "A" resets failed attempt counter and clears all alerts → idle state
- Pressing "D" displays access log on LCD (last 3 access attempts with time)
- Pressing "C" exits admin mode → idle state
- If any sensor is triggered while in this state, transition to monitoring state.

3. LCD Display Behavior

The LCD should always display the current state of the system on the first line.
• When the user enters the first digit of their code in the idle state, the second line should display left-aligned "Code: * _ _ _ _" with the "_" characters being replaced with "*" for each successive digit entered.
• If the user presses the "C" key, then the last entered character should be deleted and replaced with "_" unless there are no characters left, in which case the display should return to showing "Enter Code".
• When all five digits have been entered, the second line should display "Press B to Submit".
• Any key other than digits (0-9), B, C, A, or D should be ignored.

4. Sensor Behavior

The system includes FOUR sensors (emulated with switches):
• Sensor 1: Door contact sensor (detects door open/close)
• Sensor 2: Motion sensor (PIR - detects movement in entrance area)
• Sensor 3: Motion sensor (PIR - detects movement in secure area)
• Sensor 4: Occupancy sensor (detects presence in room)
When any sensor is active (circuit broken), the corresponding LED should illuminate.

5. Additional Requirements

• User Access Code: 5 digits (you choose, e.g., 12345)
• Admin Access Code: 5 digits, different from user code (you choose, e.g., 99999)
• Failed Attempt Tracking: System must count consecutive failed attempts
• Timing Requirements:
- Code entry timeout: 30 seconds
- Access granted display: 10 seconds
- Door open timeout: 15 seconds

- Access denied display: 5 seconds
- Lockdown duration: 2 minutes
- Monitoring auto-clear: 30 seconds of sensor inactivity

6. Implementation Notes

• Use Timer objects for all timing requirements
• Implement proper debouncing for keypad inputs (50ms)
• Use state machine pattern with clear state transitions
• Store codes as arrays or strings
• Consider using interrupts for sensor monitoring
• Implement proper code comparison logic
• Use enumerated types for states (enum State {IDLE, CODE_ENTRY, ...})

7. Testing Scenarios You Should Demonstrate

1. Successful access: Enter correct code → Green LED → Door opens → Return to idle
2. Failed access: Enter wrong code → Red LED blinks → Show attempt count → Return to idle
3. Lockdown trigger: Three wrong codes → Lockdown for 2 minutes
4. Admin override: During lockdown, enter admin code → Admin mode → Reset → Idle
5. Code deletion: Enter partial code → Press C to delete → Continue entering
6. Sensor monitoring: Trigger sensor in idle → Monitoring state → Enter code to clear
7. Door held open: Grant access → Open door → Keep open >15 seconds → Monitoring alert

--- END OF SPECIFICATION ---